

**GROUP WORK PROJECT #1**  
**GROUP NUMBER: 15588**

MScFE 610: FINANCIAL ECONOMETRICS

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Question 1a.

No, the unbiasedness is, in general violated.

The true model is  $y_i = \alpha + \beta x_i + \gamma w_i + \delta z_i + \varepsilon_i$  with the standard assumptions  $E(\varepsilon_i | x_i, w_i, z_i) = 0$ , while in the regression, where one variable is omitted, the impact of the omitted variable is integrated into the error term. The error term now becomes  $\mu_i = \delta z_i + \varepsilon_i$ .

To test the assumptions, we need  $E(\mu | X, W) = 0$ .

However, the conditional mean of  $E(\mu | X, W)$  now translates to the conditional mean of  $\delta z$  equals zero given  $X, W, Z$  are independent of each other. Unless we have  $\delta$  equals to zero or the mean of  $Z$  equals zero, the new error term  $\mu$  is in general biased.

Question 1b.

$$\beta^\Lambda = \beta + \frac{\delta X'Z}{X'X} + \frac{\delta X'\varepsilon}{X'X}$$
$$\gamma^\Lambda = \gamma + \frac{\delta W'Z}{W'W} + \frac{\delta W'\varepsilon}{W'W}$$

Question 1c.

Since the conditional mean of the true error term is zero so  $E(\beta^\Lambda) = \beta + E(\frac{\delta X'Z}{X'X})$  and since we only have orthogonal  $X$  and  $W$ , nothing is known about the relationship with  $Z$ . If  $X$  is independent from  $Z$  or if  $\delta$  is zero,  $\beta^\Lambda$  is not biased. Same conditions for  $\gamma^\Lambda$ .

Question 1d.

Two possibilities: 1. if true  $\gamma$  is equal to 0, the omitted variable has no impact on the regressed model. 2. If  $Z$  is independent from  $X, W$ , the omitted variable will not bias the estimated parameters and its impact will be absorbed in the new unbiased error term.

Question 1e.

```
import numpy as np
import statsmodels.api as sm

np.random.seed(123)

n = 500
```

```
# True parameters
a = 2
b = 3
c = 4

# Correlated X and Z
X = np.random.normal(size=n)
Z = 0.7*X + np.random.normal(size=n)
e = np.random.normal(size=n)
Y = a + b*X + c*Z + e

#####
# Correct model
#####

X_full = sm.add_constant(np.column_stack((X,Z)))
model_full = sm.OLS(Y, X_full).fit()

#####
# Omitted variable model
#####

X_omit = sm.add_constant(X)
model_omit = sm.OLS(Y, X_omit).fit()
```

## OLS Regression Results

```

=====
Dep. Variable:          y      R-squared:          0.983
Model:                  OLS    Adj. R-squared:       0.983
Method:                  Least Squares    F-statistic:      1.481e+04
Date:                    Mon, 29 Jun 2026    Prob (F-statistic): 0.00
Time:                    22:40:32    Log-Likelihood:   -662.47
No. Observations:        500    AIC:              1331.
Df Residuals:            497    BIC:              1344.
Df Model:                 2
Covariance Type:         nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const         2.0570      0.041     50.300      0.000        1.977        2.137
x1             2.8930      0.050     57.874      0.000        2.795        2.991
x2             4.0230      0.041     98.361      0.000        3.943        4.103
=====

```

```

=====
Omnibus:          0.054    Durbin-Watson:      2.024
Prob(Omnibus):    0.973    Jarque-Bera (JB):    0.134
Skew:             -0.007    Prob(JB):            0.935
Kurtosis:         2.921    Cond. No.            2.01
=====

```

## OLS Regression Results

```

=====
Dep. Variable:          y      R-squared:          0.662
Model:                  OLS    Adj. R-squared:       0.662
Method:                  Least Squares    F-statistic:      976.3
Date:                    Mon, 29 Jun 2026    Prob (F-statistic): 1.89e-119
Time:                    22:40:53    Log-Likelihood:   -1417.2
No. Observations:        500    AIC:              2838.
Df Residuals:            498    BIC:              2847.
Df Model:                 1
Covariance Type:         nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const         1.8956      0.185     10.265      0.000        1.533        2.258
x1             5.7475      0.184     31.245      0.000        5.386        6.109
=====

```

```

=====
Omnibus:          0.126    Durbin-Watson:      1.991
Prob(Omnibus):    0.939    Jarque-Bera (JB):    0.161
Skew:             -0.038    Prob(JB):            0.923
Kurtosis:         2.956    Cond. No.            1.04
=====

```

Simulation shows that omitting a relevant, correlated predictor biases the estimated coefficient of the included predictor. Increasing the sample size reduces variance but does not remove the bias; the estimator converges to the biased probability limit rather than the true parameter.

#### Question 2a.

Ordinary Least Squares (OLS) regression estimates model parameters by minimizing the sum of squared residuals. Because the residuals are squared, observations with unusually large errors receive disproportionately large weights. Consequently, OLS is highly sensitive to outliers.

There are two principal types of outliers:

A vertical outlier has an unusual response value  $y$  but a relatively typical predictor value  $x$ . Such observations produce large residuals, increasing the residual sum of squares. Depending on their location, they may noticeably change the intercept and slope estimates, although their influence is often moderate if they do not have high leverage.

A high-leverage observation has an extreme value of the predictor  $x$ . Since OLS attempts to minimize squared residuals, observations far from the center of the predictor distribution can exert substantial influence on the fitted regression line. If a high-leverage point also has a large residual, it may dramatically alter the estimated slope.

The influence of an observation depends on both its leverage and its residual. High leverage alone does not necessarily imply high influence; if the observation follows the underlying linear relationship, it may actually improve estimation. Conversely, a high-leverage point that deviates from the true relationship can dominate the estimation.

For these reasons, practitioners routinely examine residual plots, leverage values, Cook's distance, and studentized residuals before interpreting regression results. When influential observations are present, robust regression techniques or careful data validation may be preferable to ordinary least squares.

#### Question 2b.

To illustrate the impact of outliers, we simulate data from a known linear model.

The true model is

$$Y = 2 + 3X + \varepsilon$$

where :  $X \sim N(0, 1)$ ,  $\varepsilon \sim N(0, 1)$  and sample size  $n=100$ ;

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Since the true slope is known, we can compare estimated coefficients before and after introducing an outlier.

```
import numpy as np

import statsmodels.api as sm

import matplotlib.pyplot as plt


np.random.seed(123)


n = 100

X = np.random.normal(size=n)

e = np.random.normal(size=n)

Y = 2 + 3*X + e

X1 = sm.add_constant(X)

model_clean = sm.OLS(Y, X1).fit()

print("Clean model parameters:" + str(model_clean.params))

X[-1] = 10

Y[-1] = 40

model_outlier = sm.OLS(Y, sm.add_constant(X)).fit()
```

```
print("Outlier model parameters:" + str(model_outlier.params))
```

```
Clean model parameters:[1.98091425 2.98340745]
```

```
Outlier model parameters:[2.01798712 3.34114968]
```

### Question 2c.

The OLS estimator for the slope is

$$\beta^{\Lambda} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Suppose that the first  $n-1$  observations are finite numbers and there is one outlier with  $x_n \rightarrow \infty$ .

Since the first  $n-1$  observations are fixed while  $x_n$  becomes arbitrarily large,  $x \sim x_n/n$  and  $y \sim y_n/n$ .

Consequently, the contribution of the outlier grows on the order of  $x_n y_n$  in the numerator and  $x_n^2$  in the denominator, whereas the contributions from the remaining  $n-1$  observations remain bounded and become negligible as  $x_n \rightarrow \infty$ .

Therefore,  $\beta^{\Lambda} = \frac{y_n}{x_n}$ .

This shows that as the leverage point moves infinitely far from the rest of the data, the OLS slope is determined almost entirely by the outlier. The remaining observations have an asymptotically negligible influence, illustrating the extreme sensitivity of OLS to high-leverage observations.

### Question 3a. Model Selection

#### Objective and data

I seek the best linear regression model for the response  $Y$  using a subset of the five candidate predictors  $X_1, X_2, X_3, X_4, X_5$ . The dataset contains  $n = 100$  complete observations. Rather than relying on trial and error, I apply six principled procedures and check whether they converge on a common model.

Agreement across methods that penalise complexity differently is strong evidence that the selected model is genuinely best, not an artefact of one rule.

### Approaches used

I use the following six selection procedures:

- Backward elimination by p-value (significance threshold 0.05): start from the full model and repeatedly drop the least significant predictor until all remaining coefficients are significant.
- Forward selection by AIC: start from the intercept-only model and add, at each step, the predictor that lowers the Akaike Information Criterion the most, stopping when no addition improves AIC.
- Backward elimination by AIC: start from the full model and remove the predictor whose removal lowers AIC the most, stopping when no removal helps.
- Best-subset search by Adjusted R-squared: fit all 31 non-empty subsets and pick the one with the highest adjusted R-squared.
- Best-subset search by AIC and by BIC: fit all 31 subsets and pick the model minimising AIC, and separately the model minimising BIC.

### The criteria, briefly

Adjusted R-squared rewards fit but penalises extra parameters, so unlike ordinary R-squared it does not mechanically rise when useless predictors are added.  $AIC = 2k - 2 \ln L$  and  $BIC = k \ln(n) - 2 \ln L$ , where  $k$  is the number of estimated parameters,  $L$  is the maximised likelihood, and  $n$  is the sample size. Both trade goodness of fit against complexity; BIC penalises each extra parameter more heavily ( $\ln n = \ln 100 \approx 4.6$  versus 2 for AIC), so it favours smaller models. Lolr AIC or BIC is better.

### The full model as a starting point

Fitting all five predictors gives R-squared = 0.649 and adjusted R-squared = 0.630. The coefficient on X1 is tiny and far from significant, which immediately flags it as a candidate for removal:

| Predictor | Coefficient | Std. error | t-stat | p-value |
|-----------|-------------|------------|--------|---------|
| Const     | 1.190       | 0.090      | 13.25  | 0.000   |
| X1        | -0.028      | 0.184      | -0.15  | 0.881   |
| X2        | -0.585      | 0.092      | -6.35  | 0.000   |
| X3        | 0.560       | 0.092      | 6.09   | 0.000   |
| X4        | 0.710       | 0.082      | 8.62   | 0.000   |
| X5        | -0.196      | 0.084      | -2.33  | 0.022   |

*X1 has  $p = 0.881$ , so it explains little once the others are present. Every other predictor is significant at the 5% level.*

### Step-by-step results of each procedure

1) Backward elimination by p-value. Step 1 drops X1 ( $p = 0.881$ , the largest p-value, above 0.05). Re-fitting on {X2, X3, X4, X5}, all four p-values are below 0.05 (the largest is X5 at 0.021), so the procedure stops.



2) Forward selection by AIC. Starting from AIC = 357.25 (intercept only), it adds X4 (AIC 325.0), then X3 (296.4), then X2 (264.3), then X5 (260.6). Trying to add X1 raises AIC to 262.6, so it stops at {X4, X3, X2, X5}.

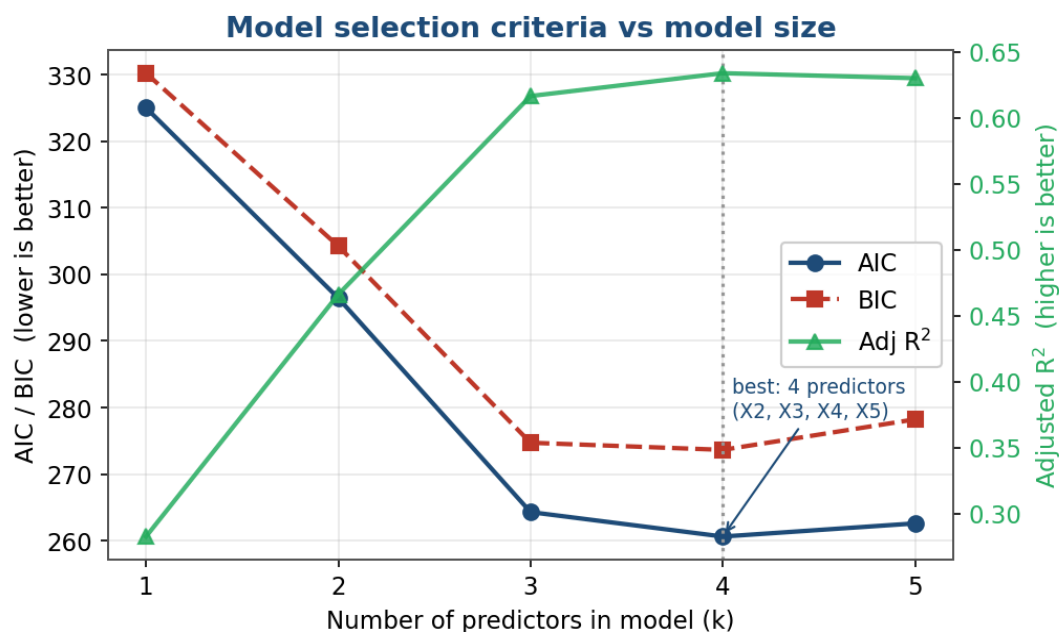
3) Backward elimination by AIC. From the full model (AIC 262.6), removing X1 lowers AIC to 260.6. No further removal lowers AIC, so it stops at {X2, X3, X4, X5}.

4-6) Best-subset search. Across all 31 subsets, the model {X2, X3, X4, X5} simultaneously maximises adjusted R-squared (0.634) and minimises both AIC (260.6) and BIC (273.6). The full five-predictor model is second on every criterion, confirming X1 adds nothing.

#### Convergence of all six methods

| Selection procedure        | Model chosen               | Key value                  |
|----------------------------|----------------------------|----------------------------|
| Backward by p-value        | $Y \sim X2 + X3 + X4 + X5$ | all $p < 0.05$             |
| Forward by AIC             | $Y \sim X2 + X3 + X4 + X5$ | AIC = 260.6                |
| Backward by AIC            | $Y \sim X2 + X3 + X4 + X5$ | AIC = 260.6                |
| Best subset, Adj R-squared | $Y \sim X2 + X3 + X4 + X5$ | Adj R <sup>2</sup> = 0.634 |
| Best subset, AIC           | $Y \sim X2 + X3 + X4 + X5$ | AIC = 260.6                |
| Best subset, BIC           | $Y \sim X2 + X3 + X4 + X5$ | BIC = 273.6                |

*All six procedures, including the most conservative (BIC) and the most permissive (adjusted R-squared), agree on the same four-predictor model. This unanimity is the central justification for the choice.*



*Figure 1. Best AIC, BIC and adjusted R-squared at each model size. AIC and BIC are both minimised at four predictors, and adjusted R-squared plateaus there; adding the fifth predictor (X1) makes every criterion worse.*

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## Final recommended model

The selected model is:

$$Y = 1.189 - 0.586 X_2 + 0.559 X_3 + 0.710 X_4 - 0.197 X_5 + \text{error}$$

| Predictor | Coefficient | Std. error | t-stat | p-value |
|-----------|-------------|------------|--------|---------|
| const     | 1.189       | 0.089      | 13.33  | 0.000   |
| X2        | -0.586      | 0.091      | -6.44  | 0.000   |
| X3        | 0.559       | 0.091      | 6.12   | 0.000   |
| X4        | 0.710       | 0.082      | 8.67   | 0.000   |
| X5        | -0.197      | 0.083      | -2.36  | 0.021   |

Fit statistics: R-squared = 0.649, adjusted R-squared = 0.634, AIC = 260.6, BIC = 273.6, F-statistic = 43.9 ( $p < 0.001$ ). Compared with the full model, dropping X1 raised the adjusted R-squared (from 0.630 to 0.634) and lowered both AIC and BIC, so the smaller model is strictly preferred on every count. All four retained predictors are significant at the 5% level.

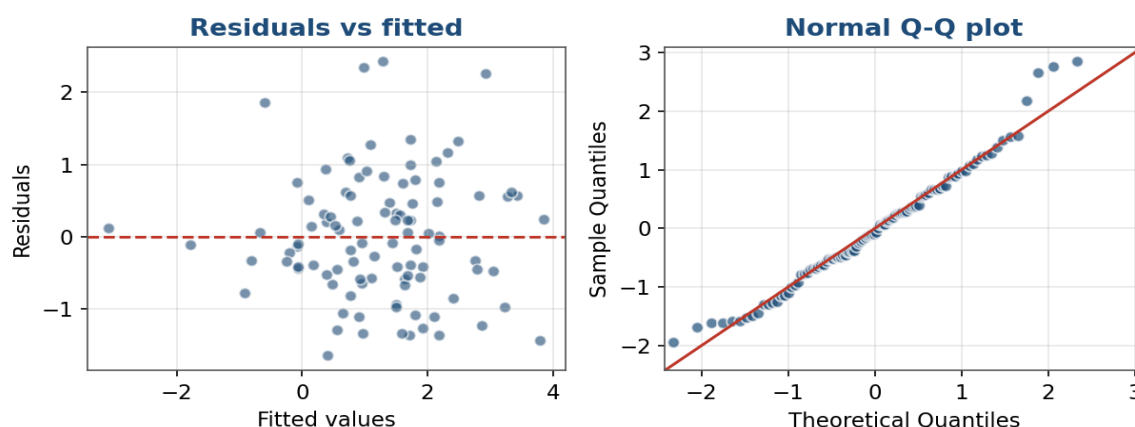
Residual diagnostics for final model  $Y \sim X_2 + X_3 + X_4 + X_5$ 

Figure 2. Residual diagnostics for the selected model. Residuals scatter randomly around zero with no obvious pattern (supporting linearity and constant variance), and the Q-Q plot lies close to the 45-degree line, consistent with approximately normal errors.

## Problem 3b. BIC versus AIC: which is more reliable?

Short answer for a client or colleague: if the goal is to recover the true model, BIC is the more reliable criterion. BIC is *consistent*, meaning that as the sample size grows it selects the true model with probability tending to one. AIC is *not* consistent: even with infinite data, it keeps a strictly positive probability of overfitting (retaining unnecessary predictors). The proof below makes this precise.

Let the true data-generating model be

$$y(i) = \alpha + \beta x(i) + \varepsilon(i),$$

but suppose I also consider a larger model that adds one redundant predictor  $z$  whose true coefficient is zero:

$$y(i) = \alpha + \beta x(i) + \gamma z(i) + \mu(i), \text{ with true } \gamma = 0.$$

The small (true) model has  $k$  parameters; the large (overfit) model has  $k + 1$ . A good criterion should reject the redundant  $z$  and pick the small model, at least asymptotically.

The likelihood-ratio difference does not grow with  $n$

Let  $LR = 2(\ln L_{\text{large}} - \ln L_{\text{small}})$  be twice the gain in maximised log-likelihood from adding  $z$ . Because  $z$  is redundant ( $\gamma = 0$  is true), standard theory says that as  $n \rightarrow \infty$  the statistic  $LR$  converges in distribution to a chi-squared variable with one degree of freedom:

$$LR \rightarrow \chi^2(1).$$

The crucial point:  $LR$  stays *bounded in probability* (it is  $O_p(1)$ ). It does not grow with  $n$ . The penalty terms, however, behave very differently, and that difference decides everything.

AIC: the penalty is too small (not consistent)

AIC adds 2 per parameter. AIC chooses the large model when its AIC is lower, that is when the fit gain beats the extra penalty of 2:

$$\text{AIC picks overfit model} \Leftrightarrow LR > 2.$$

Since  $LR \rightarrow \chi^2(1)$ , the probability of wrongly choosing the larger model tends to

$$P(\chi^2(1) > 2) \approx 0.157.$$

This limit is a fixed positive number, roughly 16%, no matter how large  $n$  becomes. So AIC retains a redundant predictor about one time in six even with unlimited data. AIC is therefore not consistent; it tends to overfit.

BIC: the penalty grows with  $n$  (consistent)

BIC adds  $\ln(n)$  per parameter. BIC chooses the large model only when

$$\text{BIC picks overfit model} \Leftrightarrow LR > \ln(n).$$

Now the threshold  $\ln(n)$  grows without bound, while  $LR$  stays  $O_p(1)$  and converges to  $\chi^2(1)$ . Therefore

$$P(LR > \ln(n)) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

The probability of keeping the redundant  $z$  vanishes. At the same time, for a genuinely needed predictor the fit gain  $LR$  grows linearly in  $n$  and easily clears the slowly growing  $\ln(n)$  hurdle, so BIC keeps the predictors that matter. Hence, BIC selects the true model with probability tending to one: it is consistent.

Conclusion and the practical caveat

Because its penalty grows with the sample size, BIC asymptotically identifies the true model with probability one, whereas AIC has a fixed, positive overfitting probability. For the purpose of selecting the correct model, BIC is the more reliable criterion.

Honest caveat: this does not make AIC useless. AIC targets a different goal, best out-of-sample prediction, and is asymptotically optimal (efficient) when the true model is very complex or not among the candidates. The two criteria answer different questions. But the question asked here, which one reliably finds the true model as  $n$  grows, has a definite answer: BIC.

#### Question 4. Interpreting the wage regression

Model:  $\ln(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{gender} + \beta_3 \text{age} + \beta_4 \text{age}^2 + \beta_5 \text{degree} + \varepsilon$ , with gender = 1 if female (0 if male) and degree = 1 for a master's degree or higher.

Estimated coefficients ( $n = 2000$ , R-squared = 0.44):

| Variable         | Coefficient | Std. error | t-stat |
|------------------|-------------|------------|--------|
| Intercept        | 0.85        | 0.12       | 7.08   |
| Educ             | 0.08        | 0.01       | 8.00   |
| gender (female)  | -0.15       | 0.10       | -1.50  |
| Age              | 0.19        | 0.06       | 3.17   |
| age <sup>2</sup> | -0.002      | 0.0003     | -6.67  |
| Degree           | 0.22        | 0.04       | 5.50   |

##### 4a. Why include age<sup>2</sup>?

Earnings do not rise at a constant rate over a career; they typically increase rapidly when a worker is young and gaining experience, then flatten and eventually decline near the end of working life. A model with age alone would force a constant marginal effect (a straight line). Adding age<sup>2</sup> lets the relationship be a curve. With  $\beta_3 > 0$  and  $\beta_4 < 0$  the curve is an inverted U, capturing diminishing returns to age and a single peak earning age. The strongly significant negative age<sup>2</sup> coefficient ( $t = -6.67$ ) confirms this concave shape is present in the data.

##### 4b. Why use $\ln(\text{wage})$ rather than wage? (statistical reasons)

There are several technical reasons:

- Skewness and normality: wage data are strongly right-skewed (a long upper tail of high earners). Taking logs compresses that tail and makes the distribution closer to normal, which improves the reliability of t-tests and confidence intervals.
- Heteroskedasticity: the spread of wages tends to grow with the level of wages. The log transform stabilises the variance, bringing the model closer to the constant-variance assumption of OLS.
- Outliers: very high salaries have far less leverage in logs, so the estimates are more robust.
- Convenient coefficients: with a log dependent variable, slope coefficients are (approximately) percentage changes, i.e. semi-elasticities, which are stable and easy to compare.

##### 4c. Why use $\ln(\text{wage})$ ? (the simple, non-mathematical reason)

Because people naturally think about pay in *percentage* terms, not in absolute currency amounts. A raise of \$2 per hour means a great deal to someone earning \$10 and almost nothing to someone earning \$200, but “a 5% raise” is meaningful and comparable for everyone. Using  $\ln(\text{wage})$  makes each coefficient read directly as a percentage effect (for example, “one more year of schooling raises pay by about 8%”), which is exactly how returns to education, experience and qualifications are usually discussed. It is the intuitive, business-friendly scale.

#### 4d. Interpretation of $\beta_1 = 0.08$

Because the dependent variable is in logs,  $\beta_1$  is a semi-elasticity. One additional year of schooling is associated with roughly an 8% increase in the hourly wage, holding gender, age and degree constant. The exact figure is  $e^{0.08} - 1 = 0.0833$ , i.e. about 8.3%, but the quick reading “about 8% per year of education” is standard. The effect is highly significant ( $t = 8.00$ ).

#### 4e. Is there a gender gap?

The point estimate  $\beta_2 = -0.15$  suggests women earn less than men: about 15% less (more precisely  $e^{(-0.15)} - 1 = -13.9\%$ ), all else equal. However, the t-statistic is only -1.50, whose absolute value is below the 5% critical value of about 1.96 (two-tailed  $p \approx 0.13$ ). I therefore cannot reject the null hypothesis of no gender effect at the 5% level: the estimated gap is not statistically significant. In plain terms, the sample hints at a gap of around 14%, but the evidence is not strong enough to conclude that a true gender gap exists at conventional significance levels. (It would, however, be significant at roughly the 13% level, so the result is suggestive rather than conclusive.)

#### 4f. At what age is hourly salary highest?

Differentiate  $\ln(\text{wage})$  with respect to age and set it to zero:

$$d \ln(\text{wage}) / d \text{age} = \beta_3 + 2 \beta_4 \cdot \text{age} = 0 \Rightarrow \text{age}^* = -\beta_3 / (2 \beta_4).$$

Substituting  $\beta_3 = 0.19$  and  $\beta_4 = -0.002$ :

$$\text{age}^* = -0.19 / (2 \times -0.002) = 0.19 / 0.004 = 47.5 \text{ years}.$$

Since  $\beta_4 < 0$  the parabola opens downward, so this is a maximum. Hourly wages are predicted to peak at about 47.5 years of age, then decline.

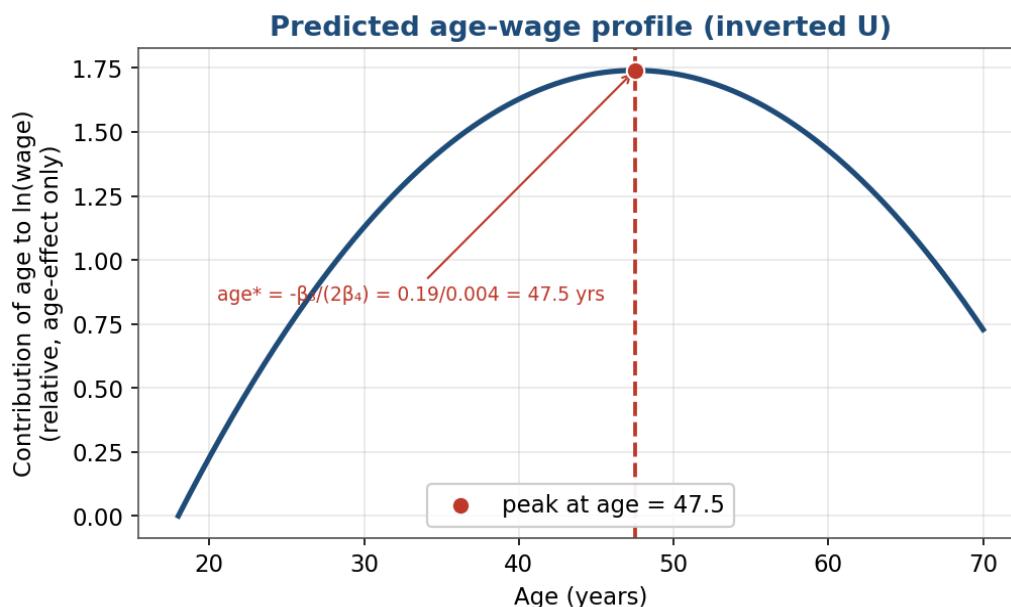


Figure 3. The estimated age effect on  $\ln(\text{wage})$  traces an inverted U that peaks at 47.5 years, after which predicted hourly wages decline.

#### 4g. Average impact of holding an advanced degree

degree is a dummy and the outcome is logged, so  $\beta_5 = 0.22$  means holding a master's degree or higher is associated with roughly a 22% higher hourly wage relative to not holding one, other things equal. The exact percentage effect for a dummy is  $e^{0.22} - 1 = 0.246$ , i.e. about 24.6%. So the advanced degree is worth approximately a 22% to 25% wage premium, and the effect is strongly significant ( $t = 5.50$ ).

#### 4h. Could I use two dummies, “male” and “female”, instead of one gender dummy?

No, not while also keeping the intercept. For every individual,  $\text{male} + \text{female} = 1$ , which is exactly the constant column already in the model. The three columns (intercept, male, female) would be perfectly collinear, the design matrix would be singular, and OLS could not estimate the coefficients. This is the ill-known dummy-variable trap. The fix is to include only  $g - 1$  dummies for a categorical variable with  $g$  categories; with two genders I include just one dummy. The omitted category (here, male) becomes the baseline absorbed into the intercept, and the single gender coefficient measures the female-versus-male difference. (One could alternatively keep both dummies but drop the intercept; that is equivalent and simply reparametrises the same model.)

5a. Random Walk Nonstationarity

Given process

$$X_t = w_t + \eta_t$$

Substitution of  $w_t = w_{t-1} + \varepsilon_t$  the recurrence relation into the process yields

$$w_{t-1} = w_{t-2} + \eta_{t-1} + \eta_t$$

Repeating until time  $t = 0$ ,

$$\begin{aligned} w_t &= w_0 + \sum_{i=0}^t \varepsilon_i \\ &= \sum_{i=0}^t \varepsilon_i \end{aligned}$$

Therefore, substituting into the original process  $X_t$ ,

$$X_t = \sum_{i=0}^t \varepsilon_i + \eta_t$$

Thus, process  $X_t$  is a random walk component with stationary white noise. This means  $X_t$ , and by association  $Y_t, Z_t$  are all nonstationary as they contain the nonstationary component  $w_t$

5b. Cointegration Vectors

The number of cointegration vectors directly corresponds to the cointegration rank  $r$ . This rank defines the number of linear combinations such that all nonstationary stochastic components of  $n$  processes are removed.

For three processes, this can be represented as

$$aX_t + bY_t + cZ_t$$

With three unknown coefficients  $a, b, c$ .

The common nonstationary trend across all processes is only  $w_t$ . Meaning the cointegration rank is

$$r = n - k = 3 - 1 = 2$$

, or in other words, two independent basis vectors are required in the coefficient space to cancel the random walk process.

Finding these vectors. Given that the processes are

$$\begin{aligned} X_t &= w_t + \eta_t \\ Y_t &= 0.5 w_t + \theta_t \end{aligned}$$

$$Z_t = -w_t + \chi_t$$

A cointegration vector is simply any vector  $\beta$  such that

$$\begin{aligned}\beta^T [X_t \ Y_t \ Z_t]^T &= aX_t + bY_t + cZ_t \\ &= a(w_t + \eta_t) + b(0.5w_t + \theta_t) + c(-w_t + \chi_t) \\ &= (a + 0.5b + c)w_t + a\eta_t + b\theta_t + c\chi_t\end{aligned}$$

Since the random walk must be removed,  $a + 0.5b + c = 0$

Rewriting in terms of  $a$  and  $b$ ,

$$\beta = A[1 \ 0 \ 1]^T + B[0 \ 1 \ 0.5]^T$$

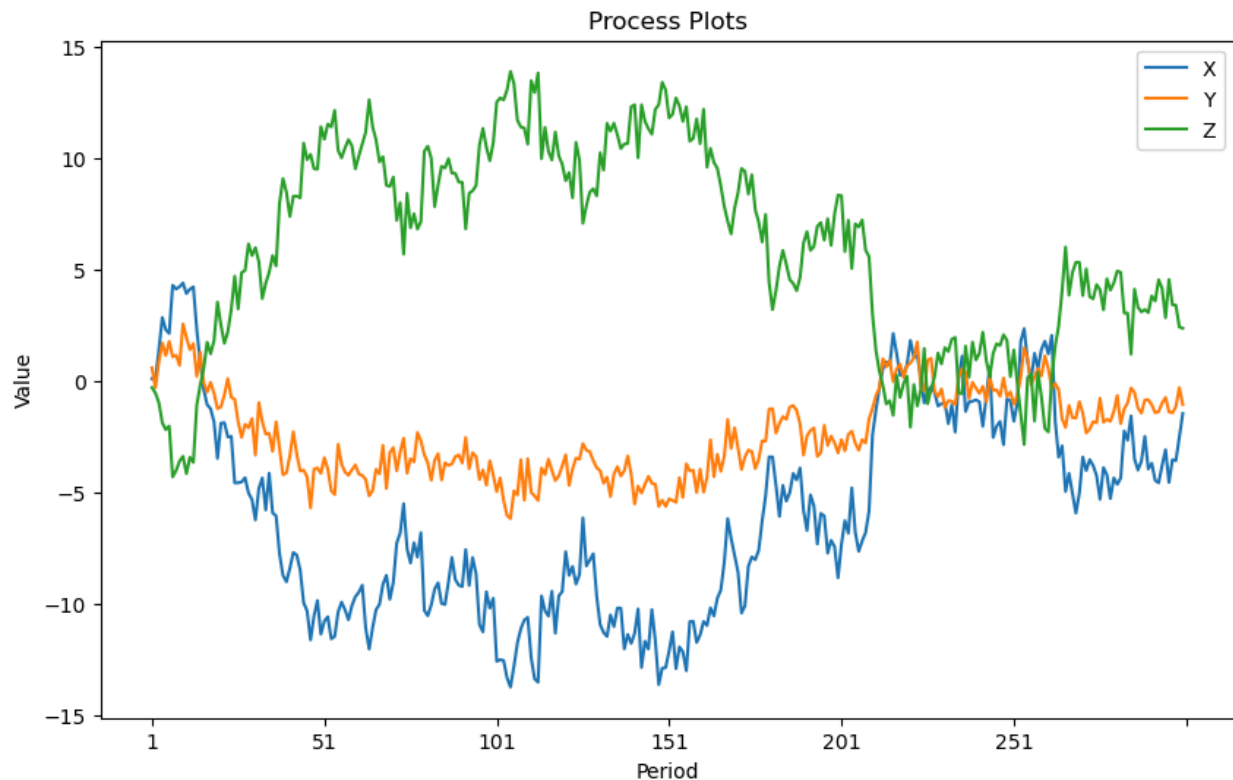
Where  $A, B$  are free parameters.

Thus,

$$\beta_1 = [1 \ 0 \ 1]^T, \quad \beta_2 = [0 \ 1 \ 0.5]^T$$

### 5c. Cointegration Testing

Visual Inspection of FE-GWP1\_dataset\_Prob5.csv time series against “Period”





Unit Root Testing can be performed to determine stationarity using the Augmented Dickey-Fuller regression test.

Assume the data satisfies:

$$\Delta Y_t = \alpha + \beta_t + \gamma Y_{t-1} + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \varepsilon_t$$

After estimating the regression, the value of the estimated coefficient  $\gamma$  for each of the regressions are

| Series | Estimated Coefficient $\gamma$ |
|--------|--------------------------------|
| $X$    | -0.041201                      |
| $Y$    | -0.066219                      |
| $Z$    | -0.038236                      |

Computing the t-statistic using the equation

$$\tau = \frac{\gamma}{SE(\gamma)}$$

Where  $SE$  is the standard error.

This statistic is compared against the Dickey-Fuller critical value, using a threshold of 5%. This value is  $c = -3.4255$ .

The null hypothesis for this test,  $H_0: \gamma \geq 0$  implies there is a unit root, and the alternative  $H_A: \gamma < 0$  is that the series is stationary.

| Series | ADF     | Outcome        |
|--------|---------|----------------|
| $X$    | -2.6430 | Fail to reject |
| $Y$    | -2.7656 | Fail to reject |
| $Z$    | -2.4535 | Fail to reject |

From the unit root test, we fail to reject the possibility that all series are stationary.

A first difference was then performed to test whether the variables are stationary after first difference, i.e.  $I(1)$

| Series     | ADF      | Outcome |
|------------|----------|---------|
| $\Delta X$ | -10.3397 | Reject  |
| $\Delta Y$ | -12.0124 | Reject  |
| $\Delta Z$ | -22.5105 | Reject  |

The Johansen test was used to determine the number of cointegrating relationships existing amongst  $n$  variables, assuming the  $X, Y, Z \sim I(1)$  prerequisite is satisfied from the ADF test.

Different lags of  $VAR_{X,Y,Z}$  models were tested, using the Akaike Information Criterion to evaluate the lag using the smallest value.

For this experiment, a lag of 6 was found to be optimal.

| Lag | AIC       |
|-----|-----------|
| 1   | -1.936352 |
| 2   | -1.919330 |
| 3   | -1.874713 |
| 4   | -1.846611 |
| 5   | -1.812227 |
| 6   | -1.790595 |
| 7   | -1.805137 |

For the Johansen test, this  $VAR_{X,Y,Z}(6)$  is then rewritten in its equivalent form  $VECM_{X,Y,X}(5)$  as shown below

$$\Delta X_t = \Pi \Delta X_{t-1} + \Gamma_1 \Delta X_{t-1} + \Gamma_2 \Delta X_{t-2} + \dots + \Gamma_5 \Delta X_{t-5} + \varepsilon_t$$

, where  $\Delta X_t$  is a vector of series  $X, Y, Z$  to estimate the long-run coefficient matrix  $\Pi$ .

Computing the values of this  $\Pi$  allows us to perform the Johansen procedure to determine its rank which is directly equivalent to the number of cointegrating relationships.

1. Maximum Eigenvalue Statistic - determines if the next largest eigenvalue  $\lambda_{r+1}$  is statistically significant

Uses the following statistic:

$$\tau = -T \ln (1 - \lambda_{r+1})$$

| $r$      | Trace   | 5% Critical Value | Decision     |
|----------|---------|-------------------|--------------|
| $= 0$    | 108.459 | 29.796            | Reject $H_0$ |
| $\leq 1$ | 53.172  | 15.494            | Reject $H_0$ |

|          |       |       |                      |
|----------|-------|-------|----------------------|
| $\leq 2$ | 3.469 | 4.598 | Fail to Reject $H_0$ |
|----------|-------|-------|----------------------|

2. Trace Test - Finds evidence for additional eigenvectors after the previous  $r$  is already assumed  
Uses the following statistic

$$\tau = -T \sum_{i=r+1}^n \ln(1 - \lambda_i)$$

This is evaluated against the null hypothesis,  $H_0$  for each rank  $r$  up to the original rank 3.

| $r$   | Maximum $\lambda$ | 5% Critical Value | Decision             |
|-------|-------------------|-------------------|----------------------|
| $= 0$ | 55.287            | 21.131            | Reject $H_0$         |
| $= 1$ | 48.574            | 14.264            | Reject $H_0$         |
| $= 2$ | 3.842             | 4.598             | Fail to Reject $H_0$ |

This implies that cointegration does exist, and there are at most two cointegrating vectors, leading to the conclusion

$$\text{rank}(\Pi) = 2$$

#### 6a. MA representation of AR Process

Given process

$$Y_t = 0.5 Y_{t-1} + \varepsilon_t$$

First check weak stationarity to ensure the AR process can be represented as an  $MA(\infty)$   
For an  $AR(1)$  process, this means

$$|\phi| < 1$$

Since  $0.5 < 1$ , this is satisfied.

Proof by substitution;

Let the process  $Y_{t-1} = 0.5 Y_{t-2} + \varepsilon_{t-1}$ , substituting into the original form yields

$$\begin{aligned} Y_t &= 0.5 (0.5 Y_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\ &= 0.5^2 Y_{t-2} + 0.5 \varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

Then process  $Y_{t-2} = 0.5 Y_{t-3} + \varepsilon_{t-2}$ , substituting,

$$\begin{aligned} Y_t &= 0.5^2(0.5 Y_{t-3} + \varepsilon_{t-2}) + 0.5\varepsilon_{t-1} + \varepsilon_t \\ &= 0.5^3 Y_{t-3} + 0.5^2 \varepsilon_{t-2} + 0.5\varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

Repeating this pattern for  $k$  substitutions:

$$\begin{aligned} Y_t &= 0.5^k Y_{t-k} + \left( \varepsilon_t + 0.5 \varepsilon_{t-1} + 0.5^2 \varepsilon_{t-2} + \dots + 0.5^{k-1} \varepsilon_{t-k+1} \right) \\ &= 0.5^k Y_{t-k} + \sum_i^{k-1} \left( 0.5^i \varepsilon_{t-i} \right) \end{aligned}$$

Taking limits  $k \rightarrow \infty$

$$Y_t = \sum_i^{\infty} \left( 0.5^i \varepsilon_{t-i} \right)$$

To which the first four terms of the representation are:  $\varepsilon_t, 0.5 \varepsilon_{t-1}, 0.25 \varepsilon_{t-2}, 0.125 \varepsilon_{t-3}$

## 6b. Autocorrelation

Given  $AR(2)$  process, the autocorrelations at lag 1 and lag 2 can be derived using the Yule-Walker equations, and satisfy the linear equation

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}$$

For order 1:

$$\begin{aligned} \rho_1 &= \phi_1 \rho_0 + \phi_2 \rho_{-1} \\ \rho_1 &= \phi_1 + \phi_2 \rho_1 \quad \text{as } \rho_0 = 1, \rho_1 = \rho_{-1} \\ \rho_1 &= \frac{\phi_1}{1 - \phi_2} \end{aligned}$$

For order 2:

$$\begin{aligned} \rho_2 &= \phi_1 \rho_1 + \phi_2 \rho_0 \\ \rho_2 &= \phi_1 \rho_1 + \phi_2 \quad \text{as } \rho_0 = 1 \end{aligned}$$

As the process is  $Z_t = 0.6Z_{t-1} + 0.2Z_{t-2} + \varepsilon_t$ , substituting for  $\phi_1 = 0.6$  and  $\phi_2 = 0.2$  yields

- $\rho_1 = 0.75$
- $\rho_2 = 0.65$